

→ If the current running sum become negative, discard it and start a new subarray from the next element

★ Kadane's Algorithm

Q53 Maximum Subarray

→ Brute force → $T.C \rightarrow O(n^2)$, $S.C \rightarrow O(1)$

~~to~~ sum $\rightarrow 0$

```
for (i = 0 to n)
{
    sum  $\rightarrow 0$ 
    for (j = 0 to n)
    {
        sum += arr[j]
        max  $\rightarrow$  Math.max(sum, max)
    }
    return max
}
```

→ Optimize → $T.C \rightarrow O(n)$ $S.C \rightarrow O(1)$

sum $\rightarrow 0$, max $\rightarrow$ Integer.MIN_VALUE

```
for (i = 0 to n)
{
    sum += arr[i]
    if (sum > max) max = sum;
}
```

if (sum < 0) {
 sum = 0
} this point cover in
 Kadane's algo

}

return max;